

学院____信息学院____

专业____计算机____

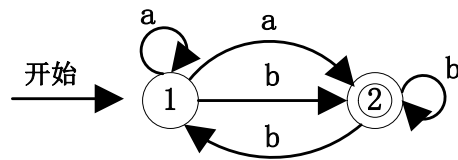
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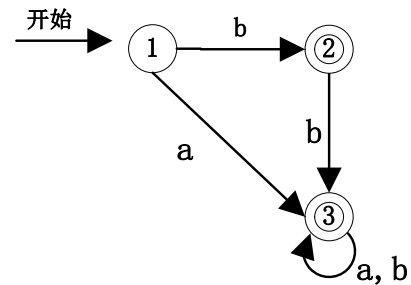
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三、计算题(每小题5分，共30分)

1. 给出与下图所示的自动机等价的确定自动机。



答:



2. 有表达式文法如下

$E \rightarrow T \mid E+T \mid$

$T \rightarrow F \mid T * F \mid$

$F \rightarrow (E) \mid i$

求表达式 $3+4*5+6$ 的规范推导

答:

$$\begin{aligned} E &\Rightarrow E+T \Rightarrow E+F \Rightarrow E+6 \Rightarrow E+T+6 \Rightarrow E+T * F+6 \\ &\Rightarrow E+T * 5+6 \Rightarrow E+F * 5+6 \Rightarrow E+4 * 5+6 \Rightarrow T+4 * 5+6 \\ &\Rightarrow F+4 * 5+6 \Rightarrow 3+4 * 5+6 \end{aligned}$$

3. 有拓广文法如下:

$E' \rightarrow$

$E \rightarrow E+T \mid$

$T \rightarrow T * F \mid F$

$F \rightarrow (E) \mid id$

如果 I 是项目集 $\{[E' \rightarrow E \bullet], [E \rightarrow E \bullet + T]\}$, 那么 $goto(I, +)$ 包括哪些项目?

答: $E \rightarrow E + \bullet T$

$T \rightarrow \bullet T * F$

$T \rightarrow \bullet F$

$F \rightarrow \bullet (E)$

$F \rightarrow \bullet id$

4. 有语法制导定义如下

$S \rightarrow E \ n \quad print(E.val)$

$E \rightarrow T \quad E.val = T.val$

$T \rightarrow F \quad T.val = F.val$

$F \rightarrow digit \quad F.val = digit.lexval$

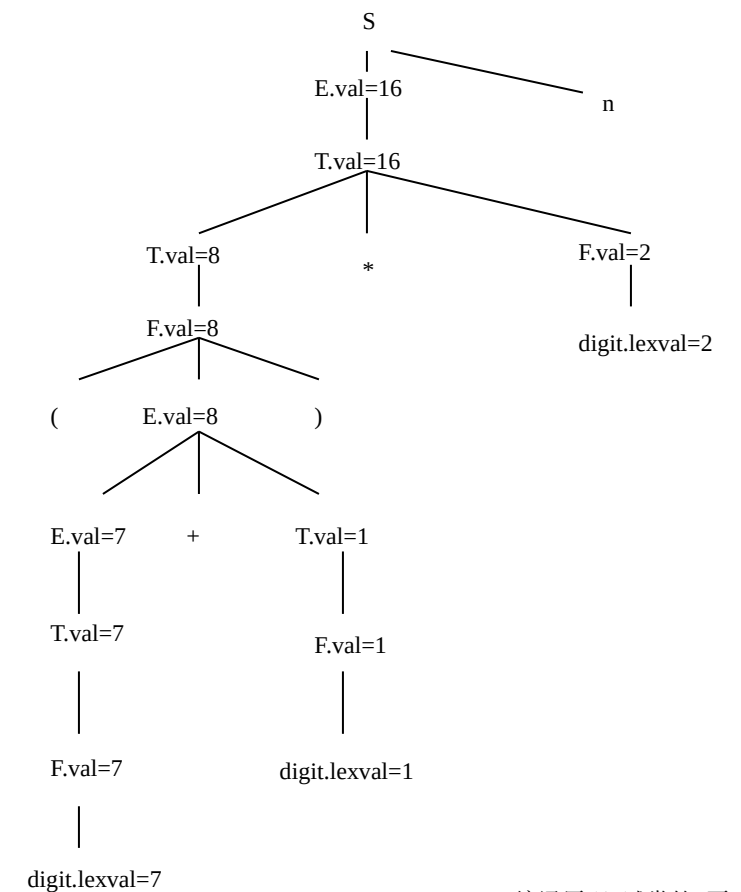
$E \rightarrow E_1 + T \quad E.val = E_1 + T.val$

$T \rightarrow T_1 * F \quad T.val = T_1.val * F.val$

$F \rightarrow (E) \quad F.val = E.val$

请为输入表达式 $(7+1)*2$ 构造注释分析树。

答:



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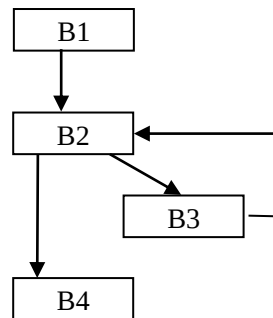
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6. 有程序如下:

```
read C
A = 0
B = 1
L1: A = A + B
    if B>C goto L2
    B = B + 1
    goto L1
L2: write A
```

对该程序划分基本块，并画出流程图。

read C A = 0 B = 1	B1
L1: A = A + B if B>C goto L2	B2
B = B + 1 goto L1	B3
L2: write A	B4



	b	(	a	)	\$
S	$S \rightarrow bMb$				
M		$M \rightarrow (L$	$M \rightarrow a$		
L		$L \rightarrow Ma)$	$L \rightarrow Ma)$		

栈	输入	产生式
\$ S	b(aa)b \$	
\$ bMb	b(aa)b \$	$S \rightarrow bMb$
\$ bM	(aa)b \$	
\$ bL(	(aa)b \$	$M \rightarrow (L$
\$ bL	aa)b \$	
\$ b)aM	aa)b \$	$L \rightarrow Ma)$
\$ b)aa	aa)b \$	$M \rightarrow a$
\$ b)a	a)b \$	
\$ b)	)b \$	
\$ b	b \$	
\$	\$	OK

[illegible]

2.构造表达式语法书的翻译方案如下

$$\begin{aligned}
 E &\rightarrow T \{R.i = T.nptr ;\} R \{E.nptr = R.s ;\} \\
 R &\rightarrow \text{addop } T \{R_1.i = \text{mknode}(\text{addop.lexeme}, R.i, T.nptr) ;\} R_1 \{R.s = R_1.s ;\} \\
 R &\rightarrow \varepsilon \{R.s = R.i ;\} \\
 T &\rightarrow (E) \{T.nptr = E.nptr ;\} \\
 T &\rightarrow \text{id} \{T.nptr = \text{mkleaf}(\text{id}, \text{id.entry}) ;\} \\
 T &\rightarrow \text{num} \{T.nptr = \text{mkleaf}(\text{num}, \text{num.val}) ;\}
 \end{aligned}$$

基于这个翻译方案构造一个预测翻译器。

答：

```

sTN *E(){
    sTN *nptr, *nptr1, *i1, *s1;
    nptr1 = T();
    i1 = nptr1;

    s1 = R(i1);
    nptr = s1;
    return nptr;}

sTN *R(sTN *i){
    sTN *nptr, *i1, *s1, *s;
    char addoplexeme;
    if ( lookahead == addop ){
        addoplexeme == lexval;
        match( addop );
        nptr = T();
        i1 = mknode ( addoplexeme, i, nptr );
        s = s1;}
    else {
        s = i;}
    return s;}

```

```

*T() {
    sTN *nptr, *nptr1, *entry;
    int val;

    if ( lookahead == '(' ) {
        match ( '(' );
        nptr1 = E();
        match ( ')' );
        nptr = nptr1; }

    else if ( lookahead == num ) {
        val = lexval;
        match ( num );
        nptr = mkleaf ( num, val ); }

    else if ( lookahead == id ) {
        entry = lexval;
        match ( id );
        nptr = mkleaf ( id, entry ); }

    else { error; }

    return nptr; }

```